

Section A

Answer **all** the questions in this section.

- 1 Enzyme X digests a sugar. Fig. 1.1 shows the rate of reaction of enzyme X and a mixture of enzyme X and substance Y on the sugar. Both reactions were carried out in a conical flask.

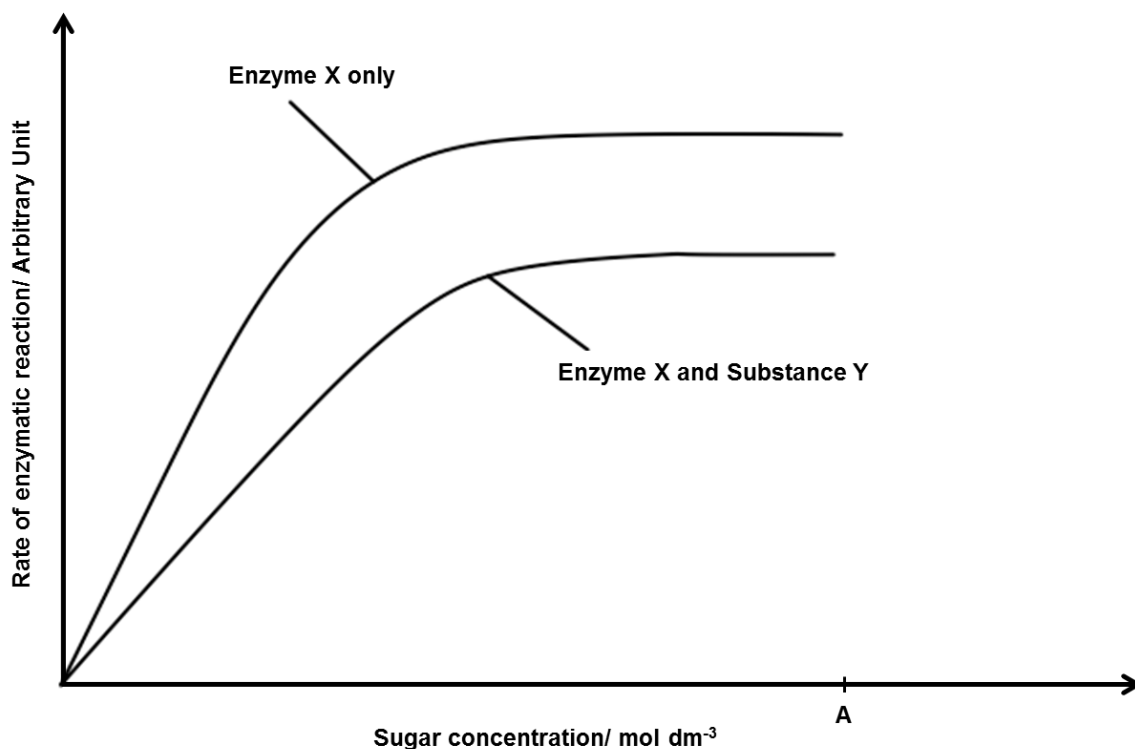


Fig. 1.1

- (a) With reference to Fig. 1.1, describe and explain the effect of substance Y on the enzymatic reaction. [3]

- Even at very high substrate concentration, the maximum rate of reaction in the presence of substance Y is lower than that in absence of substance Y/ Increasing substrate concentration cannot overcome inhibition by substance Y/ Maximum rate of reaction $V_{\max}$ is lower and affinity for the sugar is not affected by substance Y/ K_m remains unchanged;
- Substance Y is a non-competitive inhibitor which has no close structural resemblance to the substrate;
- binds irreversibly to a region other than active site of enzyme X and alters the three-dimensional conformation/ configuration of the active site such that the sugar substrate is unable to bind;

[1mark each]

- (b) Substance **Y** was found in one of the products of the reaction catalysed by enzyme **X** after the substrate concentration indicated by **A**, in Fig.1.1.

Predict how the trend of the graph for enzyme X only would continue after sugar concentrations greater than **A**. Sketch the part of the graph in Fig. 1.1. [1]

- Decreasing trend due to end-product inhibition (refer to Fig. 1).

Enzyme **X** may be immobilised in alginate beads. Fig. 1.2 shows a comparison between the activity of free enzyme **X** and immobilised enzyme **X** over a range of temperatures. Equal concentrations of free enzyme **X** and immobilised enzyme **X** were used.

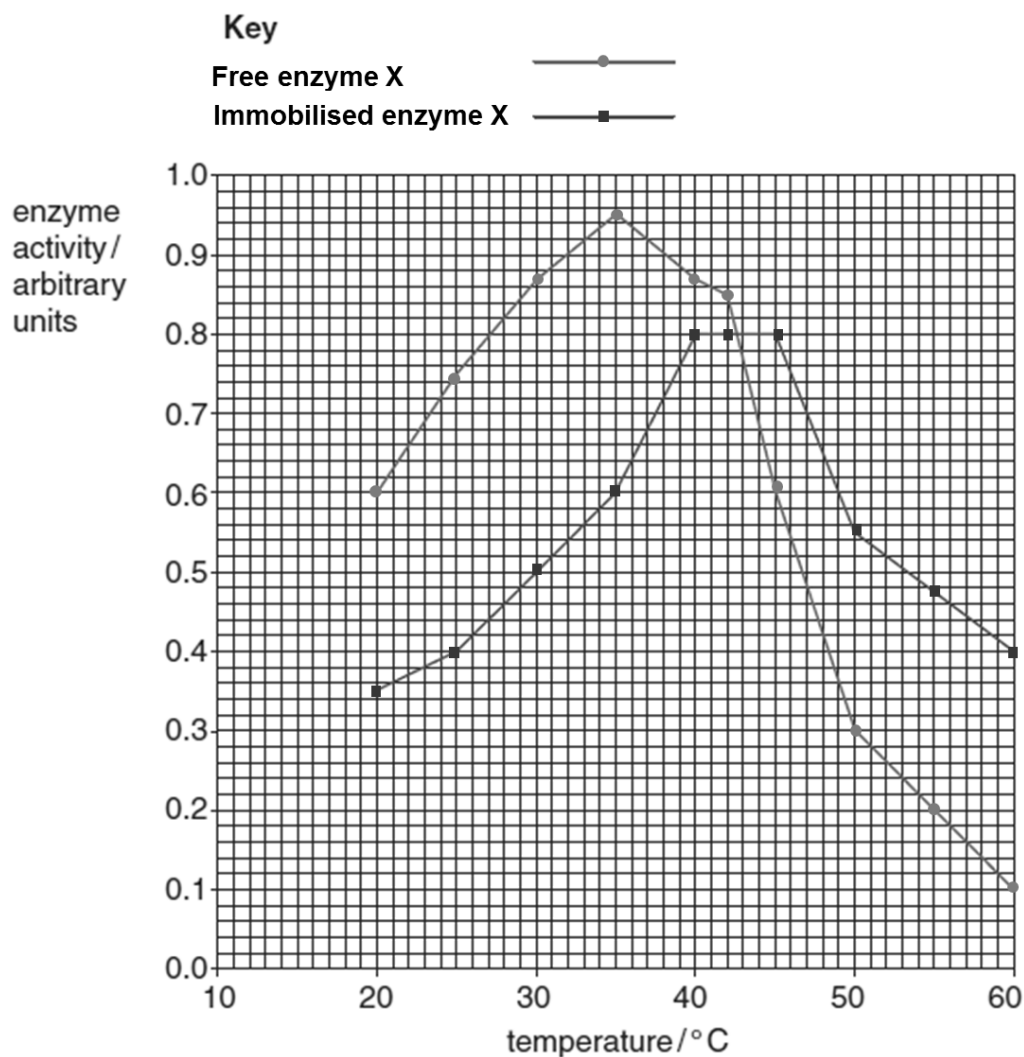


Fig. 1.2

(c) With reference to Fig. 1.2,

(i) describe and explain the effect of temperature on the activity of free enzyme **X**. [3]

- As temperature increases from 20 to 35 °C, the rate of reaction of free enzyme **X** increases from 0.60 to 0.95 A.U. due to the increased kinetic energy of the sugar substrate and enzyme molecules;
- These molecules move faster colliding with one another in correct orientation to form more **enzyme-substrate complexes**;
- The increased rate of reaction increases until it reaches its maximum rate of 0.95 A.U. at the optimum temperature of 35 °C when most enzyme-substrate complexes formed;
- Beyond the optimum temperature of 35 °C to 60 °C, rate of reaction decreases from 0.95 to 0.1 A.U. as intramolecular bonds (reference to hydrogen bonds, hydrophobic interactions) are broken and three-dimensional conformation of active sites in enzymes is altered such that substrate can no longer fit

[1 mark each, max 3 marks]

(ii) describe and explain the differences between the activity of free enzyme **X** and immobilised enzyme **X** up to 40 °C. [2]

Description

- From 20 to 40 °C, activity of immobilised enzyme is **lower** than free enzymes (reference to any suitable data range)
- Maximum activity of immobilised enzyme (0.80 A.U at 35 °C) is **lower** than that of free enzyme (0.95 A.U. at 40°C/ between 40 to 45 °C);

[1 mark each, max 1 mark]

Explanation

- Enzymes immobilised in alginate beads diffuse out from beads less easily to come into contact with sugar substrate/ less able to move/ less kinetic energy and collide less successfully with substrates leading to fewer enzyme-substrate complexes formed;
- Products formed within the beads diffuse out less easily and accumulate leading to end-product inhibition.
- Reference to thermal insulation by alginate beads so immobilised enzymes can tolerate higher temperature
- AVP

[1 mark each, max 1 mark]

(iii) suggest an advantage of using immobilised enzymes instead of free enzymes. [1]

- Immobilised enzymes have greater stability at higher temperatures/ thermostable (inference from graph);
- Immobilised enzymes have higher stability at extreme pH;
- Greater cost effectiveness as immobilised enzymes can be reused;
- Easier to purify products/ less contamination of products contained within the beads;
- AVP

[1 mark each, max 1 mark]

[Total: 10]

2 (a) Explain what is meant by homologous pairs of chromosomes. [2]

- same/ similar length/ size/ shape/ centromere location with same gene loci;
- may have different alleles for the same gene/loci;
- one of paternal origin, the other of maternal origin which pair up during prophase I

[1 mark each, max 2 marks]

(b) A mutant allele for eye colour was inherited in a recessive manner along with the X chromosome in the common fruit fly *Drosophila melanogaster*.

A red-eyed female fly was crossed with a white-eyed male fly and all the F_1 generation flies had red eyes. When a red-eyed female from the F_1 generation is crossed with a red-eyed male from the F_1 generation, all female flies had red eyes in the F_2 generation. Half of the male flies in the F_2 generation had white eyes and the rest of the males had red eyes.

- (i) Using suitable symbols, draw a genetic diagram to explain the results of this cross. [4]

Parental phenotype	red-eyed female	X	white-eyed male		
Parental genotype	$X^R X^R$	X	$X^r Y$		
Gametes	X^R		X^r	Y	[1] 1gg1
F ₁ genotype	$X^R X^r$		$X^R Y$		
F ₁ phenotype	red-eyed female		red-eyed male		[1]
F ₁ phenotype	red-eyed female	X	red-eyed male		
F ₁ genotype	$X^R X^r$	X	$X^R Y$		
Gametes	X^R	X^r	X^R	Y	[1]
F ₂ genotype	$X^R X^R$	$X^R X^r$	$X^R Y$	$X^r Y$	
F ₂ phenotype	red-eyed female	red-eyed female	red-eyed male	white-eyed male	[1]

- (ii) Explain how it can be deduced from the results of the cross that the mode of inheritance for eye colour is X-linked recessive. [2]

- when red-eyed flies crossed with white-eyed flies, progeny was all red-eyed, therefore white eye colour is recessive trait;
- when red-eyed female carrier was crossed with a red-eyed male, only the male offspring had white eyes / are affected by white eye colour, allele for white-eyed colour located on X chromosome;

[1 mark each]

- (c) Observed results of the above genetic cross differ from the expected results.

Suggest **two** reasons why such a discrepancy occurs, referring only to events that occur after meiosis. [2]

- Sample size is too small
- Variation is due to chance/ insignificant
- Differential survival of gametes/ non-random mating;
- Differential survival of fertilised zygotes/ some individuals die before being sampled

[1 mark each, max 2 marks]

[Total: 10]

- 3 An experiment was conducted to investigate the relationship between oxygen consumption and carbohydrate catabolism. A suspension of isolated mitochondria was prepared from rat liver and an intermediate of the Krebs cycle, succinic acid, was added. The reaction mixture was immediately placed in a reaction chamber of an electronic device which measured the changes in oxygen concentration in the reaction mixture. Readings were taken at 2-minute intervals for 12 minutes.

A chemical X was added to the reaction mixture 6 minutes after the start of the measurement. The experiment was repeated with another similarly prepared suspension of liver mitochondria but with no other chemicals added to serve as a control.

The results are shown in Fig. 3.

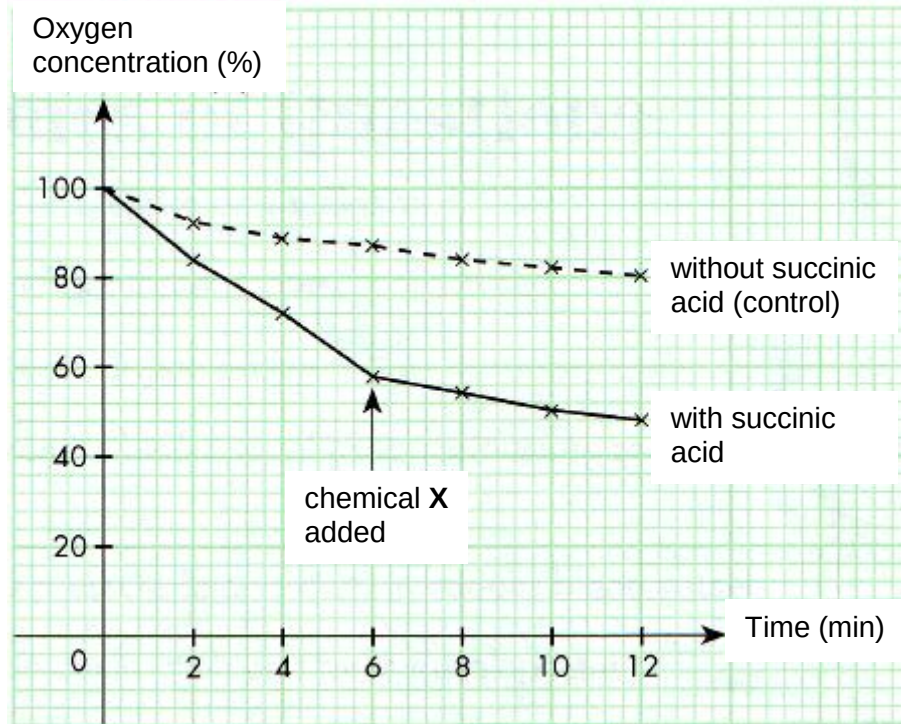


Fig. 3

- (a) Why was liver used to prepare the mitochondrial suspension but not other tissues? [2]

- The liver contains the greatest number of mitochondria in its cells [1]
 - to supply huge amounts of ATP to meet its high metabolic needs. [1]
- OR
- The suspension obtained will contain a very high density of mitochondria [1]
 - which will give the highest rate of respiration / oxygen consumption. [1]

Total: 2 max

(b) Why are carbohydrate catabolism and oxygen consumption linked? [3]

- Carbohydrate catabolism produces electrons and H^+ , which are transferred to the coenzymes FAD and NAD^+ to form $FADH_2$ and NADH respectively; [1]
- $FADH_2$ and NADH donate their electrons to the electron carriers in the electron transport chain while the H^+ remains in the matrix. As electrons travel down the electron transport chain, it is coupled to the pumping of H^+ from the matrix into the intermembrane space, via some of the electron carriers of the chain; [1]
- As H^+ flows down its concentration gradient through the ATP synthase complex in the inner mitochondrial membrane, ADP is phosphorylated to form ATP. [1]
- Molecular oxygen act as a final electron acceptor / hydrogen acceptor of the electron transport chain, forming water [1]
- so that the flow of electrons can be maintained. [1]
- So, oxygen is consumed when carbohydrate is metabolised [1].

Total: 3 max

(c) Account for the change in oxygen concentration in the control. [3]

- Residual intermediates of the Krebs cycle are metabolized continuously to release electrons / hydrogen atoms [1];
- The electrons / H^+ are transported to electron carrier system [1]
- to yield ATP using molecular oxygen as the final electron / hydrogen acceptor; [1]
- As a result, oxygen drops progressively [1].

Total: 3 max

(d) How are the changes in oxygen concentration in the reaction mixture different from those in the control from 0 to 6 minutes? [1]

- Oxygen concentration decreases more rapidly than the control [1].

OR

The oxygen concentration dropped from 100% to ~ 58% / there was a 42% drop in oxygen concentration in the reaction mixture compared to a 12 % drop in oxygen concentration / the oxygen concentration dropped from 100% to ~ 88% in the control from 0 to 6 minutes.

(e) Account for the difference mentioned in (d). [3]

- Shows reaction mixture takes up more molecular oxygen and carries out faster oxidative phosphorylation / electron transport chain. [1]
- Intermediate / succinic acid is added to reaction mixture (whereas no succinic acid added in control) (in addition to existing intermediate / succinic acid already present in isolated mitochondria before the start of the experiment; [1]
 - ✓ more electrons / H^+ will be produced; [1]
 - ✓ more input / more electrons transferred into the electron transport chain; [1]

Total: 3 max

(f) State the effect of adding chemical X to the reaction mixture on the changes in oxygen concentration. Suggest a possible role of chemical X. [2]

- Effect of adding chemical X: oxygen concentration decreases less rapidly;

OR

There was a 10% drop in oxygen concentration from ~ 58% to 48% **after** adding chemical at 6 minutes, compared to a 42% drop in oxygen concentration from 100% to ~ 58% **before** the addition of chemical X. [1]

- X = inhibitor of enzyme in Krebs cycle [1]

[Total: 14]

- 4 On the Galapagos islands, there are 14 separate species of finch. When the finches' DNA was studied, it was found that all 14 species seemed to be more closely related to a single finch species in South America than to any other bird. It is suggested that a few of these South American finches were accidentally blown onto the islands at a time before there were other birds there.

Scientists have been studying finches on one of the Galapagos islands for over thirty years.

Before 1982, the only type of ground finch living on the island was the medium ground finch, *Geospiza fortis*.

In 1982, the large ground finch, *Geospiza magnirostris* settled on the island. The large ground finch is almost twice the size of the medium ground finch.

The key food for both species is the large seed from the plant *Tribulus cistoides*. Both species use their beaks to crack open the seeds.

During the period of the study, there have been two major droughts in 1977 and 2003. Some of the observations made by the scientists were listed below:

- The drought in 1977 wiped out mainly plant species producing small seeds, leaving *Tribulus cistoides* plants that produced large seeds with hard coats.
- Following the 1977 drought, the population of the medium ground finch was greatly reduced and the average beak size had increased in both length and depth by approximately 4%.
- The 2003 drought greatly reduced seed production by all plants including *Tribulus cistoides*.

Measurements were made of the mean beak lengths and mean beak depths of the medium ground finch population in both 2002 and 2005. These are shown in Table 4.1.

Table 4.1

	2002	2005
Mean beak length / mm	11.2	10.6
Mean beak depth / mm	9.4	8.6

- (a) State the **two** main selection pressures acting on the medium ground finch population following the 2003 drought. [2]

1. food / seed shortage / competition for food;
 2. competition from the large ground finch for food
- Reject: ref. to competition without the specifics.

(b) Explain how the change in mean beak size in the medium ground finch population between 2002 and 2005 supports Darwin's theory of evolution. [4]

- There is variation in beak size in the medium ground finch population;
- presence of small seeds after the 2003 drought;
- Finches with small beaks are at a selective advantage and are more likely to survive till sexual maturity to reproduce / and have greater reproductive success;
- and pass on the advantageous allele for small beak to their offspring;
- Over a number of generations, there is an increase in allele frequency for small beak in the medium ground finch population, thus explaining the drop in average beak length and depth to 10.6 mm and 8.6 mm respectively (from 11.2mm and 9.4 mm respectively, in 2002), in 2005.

[1m each, max 4]

[Total: 6]

Section B

Answer EITHER 5 OR 6.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where applicable.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

Either

- 5 (a)** Describe the main structural features of endoplasmic reticulum and [7]
mitochondria.

- 1. There are two types of endoplasmic reticulum: rough endoplasmic reticulum (RER) and smooth endoplasmic reticulum (SER);**

Structural features of RER

- 2. Consists of network of sheets;**
- 3. Ribosomes are present on the surface of RER;**
- 4. ER membrane is continuous with the outer membrane of the nuclear envelope;**
- 5. The space between the membranes of the nuclear envelope is continuous with the cisternal space of the ER;**

Structural features of SER

- 6. Consists of network of tubules;**
- 7. No ribosomes are present on the surface of SER;**

Structural features of Mitochondria

- 8. Cylindrical/rod shape;**
- 9. Width ranges from 0.5-1.5µm while their length range from 3-10µm;**
- 10. Enclosed by double membranes/outer membrane and inner membrane;**
- 11. Separated by an extremely narrow fluid-filled space – intermembrane space;**
- 12. The outer membrane is smooth;**
- 13. The inner membrane is highly folded;**
- 14. to form cristae;**
- 15. Cristae allows the attachment of enzymes, electron transport chains, ATPase;**
- 16. The interior contains an organic, semifluid matrix;**
- 17. with (70S) ribosomes, circular DNA, and enzymes;**

[max 7]

- (b)** Outline the functions of endoplasmic reticulum and mitochondria. [5]

Functions of rER

1. Site of protein synthesis;
2. Proteins destined for secretion/for incorporation into membranes;
3. are synthesised by ribosomes attached to the rER;
4. In the ER lumen, proteins may be modified;
5. by enzymes;
6. that add complex carbohydrates or lipids to them;
7. It serves as intracellular transport system;
8. To isolate and transport the proteins synthesised to other compartments within the cell;
9. by transport vesicles budding off the ER membrane;

Functions of sER

10. Synthesis of lipids;
11. Involved in carbohydrate metabolism/glycogenolysis (in liver);
12. by removing phosphate from glucose phosphate so that the glucose can leave the cell;
13. Involved in detoxification of drugs/other poisons (in liver);
14. contain detoxifying enzymes that adds hydroxyl groups to drugs;
15. making the drugs soluble and easier to flush out of body;
16. Stores calcium ions necessary for muscle contractions;

[Points 1-16: max 4]

Functions of Mitochondria

17. Sites of aerobic respiration;
18. to synthesise ATP (using energy released from the oxidation of organic acids);
19. Matrix contains enzymes/is the site for Krebs cycle to occur;
20. Inner membrane is the site for Oxidative Phosphorylation;
21. Heat production/energy released as heat;

[Points 17-21: max 1]

[Total: 20]

- 6 (a) Compare the structure and function of *Taq* DNA polymerase and RNA polymerase. [7]

	<i>Taq</i> DNA polymerase	RNA polymerase
Similarities:		
1. Template	Both use DNA as template	
2. Structure	Both are proteins/enzymes	
Differences:		
3. Product	DNA	RNA
4. Substrate	Uses deoxyribonucleotides as substrates	Uses ribonucleotides as substrates
5. Template	Uses both DNA strands as template	Transcribes only 1 of the 2 DNA strands
6. Molecular structure	Large no. of disulphide bonds to allow <i>Taq</i> DNA polymerase to withstand high temperature	Does not have large no. of disulphide bonds, hence, denatures at high temperature
7.	Requires the presence of free 3'OH group to add the deoxyribonucleotides	Does not requires the presence of free 3'OH group to add the ribonucleotides
8. Functions	Involves in DNA replication/ PCR	Involves in protein synthesis

- (b) Outline how restriction enzymes are used for removing sections of DNA from a chromosome. [5]

- Each restriction enzyme recognizes a **specific sequence of 4 – 8 nucleotide bases** on the DNA molecule/ **restriction site**.
- Restriction enzyme has **specific active site** that can only **bind with the complementary shape of restriction site**
- The restriction enzyme acts on the restriction site by **breaking the phosphodiester bonds** at a position between 2 specific bases
- cutting up DNA molecules into **restriction fragments**.
- Restriction enzymes produce restriction fragments with **sticky ends or blunt ends**

(c) Explain how gel electrophoresis can be used to provide evidence of molecular homology. [8]

1. Gel electrophoresis can be used in **DNA analysis** to provide evidence of molecular homology.
2. **Different samples of DNA from different species** are taken and **digested/hydrolysed with a restriction enzyme**
3. that breaks the phosphodiester bonds **at specific restriction sites**.
4. Species that **differ genetically** will produce **DNA fragments of different length/size** and vice versa / are **genetically similar** will produce **DNA fragments of similar length/sizes**.
5. The DNA fragments are **loaded into the gel** and subjected to an **electric field**.
6. As DNA is **negatively charged**, the fragments will move towards the anode
7. and they will be **separated according to molecular size** as the **longer fragments will travel more slowly** through the gel/ the **shorter fragments will travel faster** through the gel.
8. The band pattern can be **visualised** by staining the gel with **ethidium bromide/other DNA staining dyes** and **viewing it under UV light**.
9. The **band pattern** produced can be **analysed / used to compare samples** from different species in order to establish molecular homology.
10. The **more similarities between 2 species** in the band pattern produced, the **more closely related the species** are and vice versa./ the **more recent they share a common ancestor**

[1 mark each, max 8]

[Total: 20]